

MULTILINGUAL EXPRESSIVE TEXT-TO-SPEECH (TTS) SYSTEM

Project Report

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Abstract

The Multilingual Expressive Text-to-Speech (TTS) System converts text into natural-sounding speech in English, Hindi, Telugu, and Kannada using React, FastAPI, Piper TTS, and gTTS. It also supports PDF and TXT to speech.

Objectives

Develop a multilingual TTS system, support four languages, convert text/PDF/TXT to speech, provide a responsive interface, and allow audio playback and download.

Technologies Used

Frontend: React.js

Backend: FastAPI

Language: Python

TTS: Piper TTS, gTTS

Libraries: Axios, PyPDF2, python-multipart

System Architecture

User → React UI → FastAPI → Piper/gTTS → WAV Audio → Play/Download

Features

Text-to-Speech, PDF-to-Speech, TXT-to-Speech, English, Hindi, Telugu, Kannada, Character Counter, Word Counter, Copy, Clear, Loading Indicator, Audio Player, Download, Responsive UI.

Implementation

React provides the UI. FastAPI processes requests and uses Piper TTS for English, Hindi, Telugu and gTTS for Kannada. Audio is returned as a WAV file.

Challenges

Limited offline Kannada models. gTTS was integrated for Kannada while Piper TTS was used for the other languages. PyPDF2 was used for PDF text extraction.

Results

The application successfully converts text, PDF, and TXT files into speech and allows users to play and download generated audio.

Conclusion

The project meets its objectives using open-source technologies and provides a simple multilingual TTS solution.

Future Enhancements

Support more Indian languages, offline Kannada models, speed and pitch control, expressive speech.

References

[React Documentation](#)

[FastAPI Documentation](#)

[Piper TTS Documentation](#)

[gTTS Documentation](#)

[PyPDF2 Documentation](#)

Appendix (web Application)

 **Multilingual Expressive TTS**
Convert text into natural sounding speech in multiple Indian languages

🗨️ **Input Text**

मेरा नाम विदुद दे ओर में प्रसिद्धी पृतिवर्ति में पद रहा हूँ

Words : 12 65/500

📋 Copy

🗑️ Clear

🌐 Language

🔊 Speech Settings

🔊 Voice

Hindi

Neutral

Default

📁 Upload PDF / TXT

📄 Choose File No file chosen

🗨️ PDF to Speech

🗨️ TXT to Speech

🔊 Generated Audio

🔊 Generate Speech

✅ Speech generated successfully!

🔊 Generated Audio

▶ 0:00 / 0:04 🔊 🔊

📄 Download Speech

📄 Speech Information

🌐 Language : Hindi

🔊 Emotion : Neutral

🔊 Voice : Default

🔊 Output : WAV Audio

🗨️ Words : 12

📄 Characters : 65

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Developed using React • FastAPI • Piper • gTTS